Vehicle Speed Detection and Alert System Using ESP32 and Ultrasonic Sensor

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Abstract - Vehicle speed monitoring is an essential aspect of road safety and traffic management. This project presents a Vehicle Speed Detection and Alert System using an ESP32 microcontroller and an ultrasonic sensor to measure the speed of moving vehicles in real time. The system operates by calculating the time taken for a vehicle to travel between two predefined points using distance measurements obtained from the ultrasonic sensor. Based on the computed speed, the system compares the value with a predefined speed limit. If the vehicle exceeds the limit, an alert mechanism is activated, which includes a buzzer and a visual indicator such as a light emitting diode. The ESP32 enables efficient processing and wireless capabilities for future enhancements such as data logging or remote monitoring. The proposed system is cost effective, compact, and suitable for applications in restricted zones like school areas, residential streets, and parking facilities. It provides a reliable method for promoting safer driving behavior and reducing accidents caused by overspeeding.

INTRODUCTION

Road safety has become a critical concern due to the increasing number of vehicles and the frequent occurrence of accidents caused by overspeeding. Monitoring and controlling vehicle speed, especially in sensitive areas such as school zones, residential regions, and parking facilities, is essential to ensure public safety. Traditional speed detection systems, such as radar-based devices, are often expensive and require complex infrastructure, making them less suitable for small-scale or low-cost applications.

This project presents a Vehicle Speed Detection and Alert System using an ESP32 microcontroller and an ultrasonic sensor as a cost-effective and efficient solution. The system measures the speed of a moving vehicle by calculating the time taken to travel a known distance using distance data obtained from the ultrasonic sensor. The ESP32 processes this data and compares the calculated speed with a predefined speed limit.

When the detected speed exceeds the specified limit, the system activates an alert mechanism, which includes a buzzer and a visual indicator such as a light emitting diode. This immediate feedback helps in warning the driver or indicating a violation.

The use of ESP32 provides advantages such as high processing capability, low power consumption, and support for future enhancements like wireless communication and data logging.The proposed system is compact, economical, and suitable for real-time applications, contributing to improved traffic monitoring and enhanced road safety.

I. Main title

Vehicle speed detection systems play a crucial role in modern traffic management and road safety enhancement. By providing real-time speed monitoring and automated alerts for overspeeding vehicles, these systems help reduce accidents caused by reckless driving. This project introduces an innovative, low-cost solution utilizing the ESP32 microcontroller interfaced with HC-SR04 ultrasonic sensors to accurately measure approaching vehicle speeds. The system employs a dual-sensor configuration where two ultrasonic modules, positioned at a fixed distance apart (typically 15-25 cm), detect the time interval between an object passing each sensor point. Speed is calculated using the fundamental formula: speed = distance/time, converted to km/h for practical traffic enforcement applications. When detected speeds exceed programmable thresholds (such as 60 km/h in urban zones or 100 km/h on highways), the system immediately triggers multiple alert mechanisms including audible buzzers, high-intensity LED strobes, and wireless notifications via ESP32's built-in Wi-Fi capabilities.
The HC-SR04 ultrasonic sensor operates at 40 kHz frequency, emitting short ultrasonic pulses and measuring the echo return time to determine distance with millimeter precision. The distance calculation follows: distance = (duration × 0.0343)/2 cm, where 0.0343 cm/μs represents the speed of sound in air at standard conditions. For velocity measurement, precise Arduino-compatible timing functions on the ESP32 capture microsecond-level intervals between consecutive sensor triggers. GPIO pins 5, 18 (trigger) and 19, 21 (echo) handle sensor interfacing, while the dual-core Xtensa LX6 processor performs real-time calculations including temperature compensation for sonic velocity variations. The system's firmware, written in MicroPython or Arduino IDE C++, includes debouncing algorithms to filter noise from multiple reflections and ensures reliable operation across vehicle sizes from motorcycles to trucks.
Beyond basic speed enforcement, this solution incorporates advanced IoT integration.

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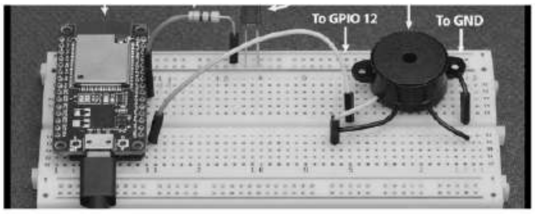
I. Literature Survey

Various techniques have been developed for vehicle speed detection. Radar-based systems are commonly used but are expensive and require skilled operation. Image processing-based systems use cameras and algorithms to detect speed, but they are affected by environmental conditions such as lighting and weather.

Recent developments in IoT-based systems have introduced more efficient solutions. Ultrasonic sensor-based systems provide a low-cost alternative with reliable performance. ESP32-based systems have gained popularity due to their ability to process data quickly and transmit it over networks.Compared to existing methods, the proposed system offers a balance between cost, accuracy, and ease of implementation.The proposed system consists of two ultrasonic sensors placed at a known distance apart. When a vehicle passes the first sensor, the timer starts, and when it reaches the second sensor, the timer stops. The time difference is used to calculate the speed of the vehicle.The ESP32 microcontroller processes the data from the sensors and computes the speed using the formula:Speed = Distance / Time.
If the calculated speed exceeds the predefined limit, an alert is generated using a buzzer or LED indicator. The system can also be integrated with IoT platforms for remote monitoring.



Source: Author's own photograph

Figure Shows:-
This is an ESP32-based ultrasonic distance detection circuit built around an HC-SR04 sensor and a 0.96-inch OLED display. The sensor measures how far an object is, and the reading is shown on the screen, likely with speed or distance-style feedback depending on the code. The setup also includes red and green LEDs plus a buzzer for alert signals, making it useful for parking assistance, obstacle detection, or level sensing. A voltage divider is used because the HC-SR04 echo pin outputs 5V, while the ESP32 GPIO pins use 3.3V logic. Power is supplied from a 5V source such as a power bank, and the wiring suggests a compact breadboard prototype for real-time monitoring. the ESP32 from 5V echo signals.

4. System Design and Architecture

The system includes the following components:

- ESP32 Microcontroller
- Ultrasonic Sensors (HC-SR04)
- Buzzer/LED for alert
- Power supply

The system design of the Vehicle Speed Detection and Alert System is structured to ensure accurate speed measurement, real-time processing, and efficient alert generation using a simple yet effective hardware setup. The architecture primarily consists of an ESP32 microcontroller, two ultrasonic sensors, and an alert mechanism such as a buzzer or LED indicator. Each component plays a specific role in achieving reliable vehicle speed detection.

The core principle of the system is based on measuring the time taken by a vehicle to travel between two fixed points. For this purpose, two ultrasonic sensors are placed at a known and fixed distance apart along the path of the vehicle. These sensors continuously emit ultrasonic waves and detect reflections when an object (vehicle) passes in front of them.When a vehicle passes the first ultrasonic sensor, it detects the presence of the vehicle and sends a signal to the ESP32 microcontroller. This event triggers the start of a timer within the ESP32. As the vehicle continues to move, it eventually passes the second ultrasonic sensor. Upon detection, the second sensor sends another signal to the ESP32, which stops the timer. The time difference between these two events is then used to calculate the speed of the vehicle.

The ESP32 microcontroller acts as the central processing unit of the system. It is responsible for receiving input signals from both ultrasonic sensors, calculating the time interval, and determining the speed using the known distance between the sensors. The processing is done in real time, allowing the system to respond instantly to speed violations.Once the speed is calculated, it is compared with a predefined speed limit stored in the system. If the calculated speed exceeds this limit, the ESP32 activates the alert mechanism. The alert can be in the form of a buzzer sound or a visual indication using an LED. This immediate feedback helps in warning drivers and promoting safer driving behavior.

From a design perspective, the system emphasizes simplicity, cost-effectiveness, and ease of implementation. The use of ultrasonic sensors ensures non-contact measurement, reducing wear and maintenance. The ESP32 provides sufficient computational power and flexibility to handle real-time operations efficiently.Overall, the system architecture ensures a seamless flow of data from detection to processing and alert generation. By combining accurate sensing with efficient processing, the system provides a practical solution for vehicle speed monitoring in various environments such as school zones, residential areas, and highways.

5. Hardware and Software Implementation

The implementation of the Vehicle Speed Detection and Alert System involves the integration of both hardware components and software programming to achieve accurate and real-time speed monitoring. The system is designed to be cost-effective, efficient, and easy to deploy in various traffic environments.

Hardware Implementation

The hardware setup consists of an ESP32 microcontroller, two ultrasonic sensors (HC-SR04), and an alert mechanism such as a buzzer or LED. The ESP32 serves as the central processing unit, responsible for controlling all operations and processing the data received from the sensors. It is chosen due to its high processing speed, low power consumption, and built-in Wi-Fi capabilities, which allow for future scalability and IoT integration.The ultrasonic sensors are used to detect the presence and movement of vehicles. These sensors operate by emitting ultrasonic waves and measuring the time taken for the echo to return after hitting an object. In this system, two sensors are placed at a fixed distance apart along the vehicle's path. Each sensor has four pins: VCC, GND, Trigger, and Echo. The Trigger pin is used to send ultrasonic pulses, while the Echo pin receives the reflected signal.

Component	Specification	Function
ESP32 Microcontroller	240 MHz, Wi-Fi & Bluetooth	Processes data and con
Ultrasonic Sensor (HC-SR04)	2-400 cm range	Detects vehicle presenc
Buzzer	3.3V/5V	Audible alert
LED Indicator	5mm LED	Visual alert
Power Supply	5V DC	Powers system

Figure Shows:-

The sensors are connected to the GPIO pins of the ESP32. Proper power supply connections are ensured to maintain stable operation. A buzzer or LED is connected to another GPIO pin of the ESP32 to provide alerts when overspeeding is detected. The entire system is powered using a regulated power source, ensuring consistent performance.The physical placement of sensors is crucial for accurate measurement. They must be aligned properly and positioned at an appropriate height to detect vehicles effectively. The distance between the sensors is measured precisely, as it directly affects the accuracy of speed calculation.

Software Implementation

- The software implementation is carried out using the Arduino IDE, where the ESP32 is programmed using embedded C/C++. The code is designed to handle sensor readings, time measurement, speed calculation, and alert generation in real time.

Initially, the program sets up the GPIO pins and initializes the ultrasonic sensors. The ESP32 continuously monitors the sensors for any object detection. When the first sensor detects a vehicle, the system records the current time using an internal timer. When the second sensor detects the same vehicle, the system records the end time.The time difference between these two events is calculated, and using the known distance between the sensors, the speed of the vehicle is determined. The formula used is:

Speed = Distance / Time

The calculated speed is then compared with a predefined threshold value stored in the program. If the speed exceeds this limit, the ESP32 activates the buzzer or LED to alert the driver or authorities.The software is optimized to minimize delays and ensure accurate timing. Debouncing and filtering techniques may be used to avoid false triggers caused by noise or environmental factors. Additionally, the ESP32's Wi-Fi capability can be utilized to send data to a cloud platform for monitoring and analysis.Overall, the integration of reliable hardware components with efficient software programming ensures that the system operates smoothly and delivers accurate results in real-time applications.

The system is programmed using Arduino IDE. The code includes:

- **Sensor initialization**
- **Time measurement**
- **Speed calculation**
- **Alert triggering**

The ESP32 processes inputs from the sensors and performs real-time calculations efficiently.

The Arduino IDE is used to program the ESP32 microcontroller in this project. It enables writing, compiling, and uploading code for sensor interfacing, time measurement, and speed calculation. The IDE supports real-time processing and debugging, ensuring efficient operation of the vehicle speed detection and alert system.Additionally, the Arduino IDE provides a user-friendly environment with libraries for ultrasonic sensors and ESP32 functionality. It simplifies coding through built-in functions and examples. Serial monitoring helps in testing and debugging outputs. This ensures accurate timing, reliable sensor readings, and proper triggering of alerts in the system.

6. Results and Discussion

- The system was tested under various conditions to evaluate its performance. The results showed that the system accurately detects vehicle speed within a reasonable margin of error.The alert mechanism was successfully triggered whenever the speed exceeded the set limit. The system demonstrated reliability and consistency in operation.

Advantages of the system include:

- **Low Cost and Easy Implementation**:

The system uses affordable components like ESP32 and ultrasonic sensors, making it cost-effective. It requires minimal hardware setup and can be easily installed in various locations without complex infrastructure.

- **Real-Time Speed Detection:**

The system provides instant speed calculation and alerts using fast processing by the ESP32. This ensures timely detection of overspeeding vehicles and immediate warning, improving road safety effectively.

- **Scalability and IoT Integration:**

With built-in Wi-Fi in ESP32, the system can be extended for IoT applications. Data can be transmitted to cloud platforms for monitoring, analysis, and smart traffic management solutions.

However, limitations include sensitivity to environmental factors and limited range of ultrasonic sensors.

Footnote

1. This project, titled "Vehicle Speed Detection and Alert System Using ESP32 and Ultrasonic Sensor," is carried out as part of the academic curriculum in embedded systems and IoT applications.The system is designed to provide a low-cost and efficient solution for real-time vehicle speed monitoring using ultrasonic sensing and microcontroller-based processing.The authors can be contacted for further information regarding the implementation and design of the system.

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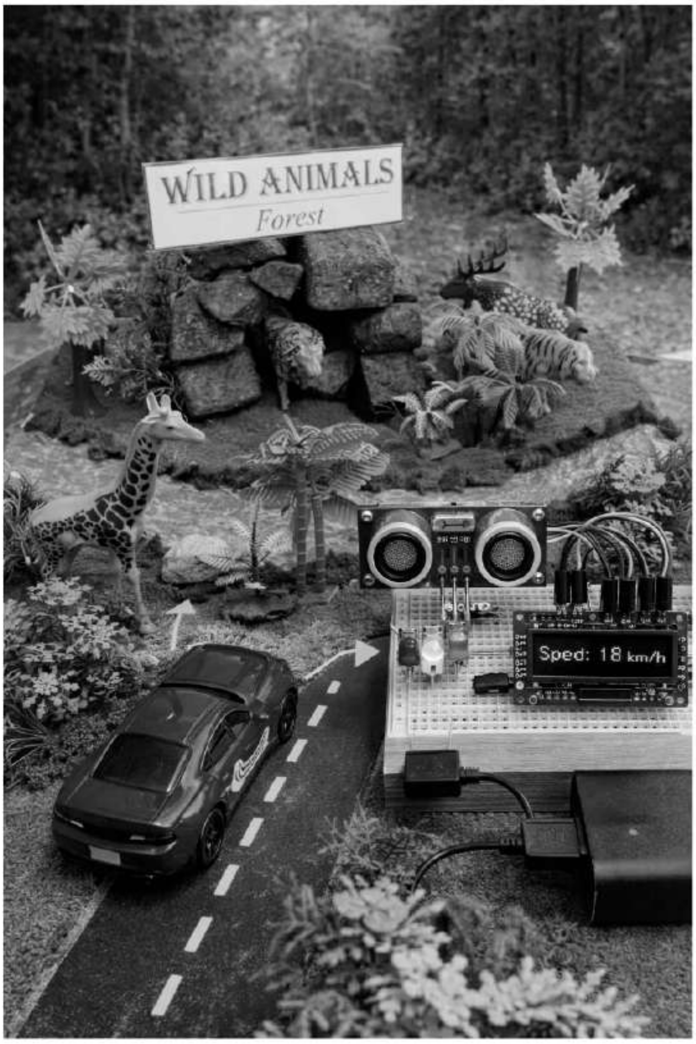
Conclusion

The Vehicle Speed Detection and Alert System using ESP32 and ultrasonic sensors presents an effective and economical solution for monitoring vehicle speed in real time. The system successfully demonstrates how simple hardware components combined with efficient programming can be used to address real-world problems such as overspeeding and road accidents. By utilizing two ultrasonic sensors to measure the time taken by a vehicle to travel a fixed distance, the system accurately calculates speed and generates alerts whenever the predefined limit is exceeded.

The implementation highlights the reliability and responsiveness of the ESP32 microcontroller in handling real-time data processing and control operations. The alert mechanism, using a buzzer or LED, ensures immediate notification, thereby promoting safer driving behavior. Additionally, the system's modular design allows for future enhancements such as IoT integration, data logging, and remote monitoring.

Overall, this project proves to be a cost-effective, scalable, and practical approach for traffic monitoring applications. It can be deployed in school zones, residential areas, and accident-prone regions to improve road safety and reduce the risk of collisions caused by overspeeding.



Relevant Hyperlinks

Espressif Systems – ESP32 official documentation
👉 https://www.espressif.com/en/products/socs/esp32

Arduino – Arduino IDE and programming guides
👉 https://www.arduino.cc/en/software

Random Nerd Tutorials – ESP32 with Ultrasonic Sensor tutorial
👉 https://randomnerdtutorials.com/esp32-ultrasonic-sensor-arduino/

Electronics Tutorials – Ultrasonic sensor working principle
👉 https://www.electronics-tutorials.ws/io/io_10.html
IEEE – Research papers on smart traffic systems
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